



BAKER'S PERCENTAGES

Formulas, especially bread formulas, are sometimes expressed in percentages called *baker's percentages*. With baker's percentages, each ingredient is expressed as a certain ratio or percent of the total amount of flour in the formula. Flour is used as the basis for baker's percentages because it is typically the predominant ingredient in most baked goods. Since the total amount of flour is designated as 100 percent, the percentages of all ingredients add up to more than 100 percent. Table 1.4 provides an example of a bread formula expressed in weight and in baker's percentages. Notice that more than one type of flour is included in this formula, but that together the weight of the flours adds up to 100 percent.

For formulas that do not contain flour, each ingredient is expressed as a percentage of the predominant and characteristic ingredient. In the case of a date filling, for example, each ingredient is expressed as a percentage of the amount of dates (Table 1.5). For baked custard, each ingredient is expressed as a percentage of dairy ingredients—milk and cream.

Baker's percentage—sometimes called *formula percentage* or indicated as “on flour weight basis”—is different than the percentages commonly taught in math classes. In the more common type of percentage, each ingredient is expressed as a certain percentage of the total batch size. In this case, ingredient percentages add up to 100 percent. Table 1.6 shows the bread formula from Table 1.4, this time expressed as a percentage of the total batch.

TABLE 1.4 ■ WHOLE WHEAT BREAD FORMULA EXPRESSED IN WEIGHT AND IN BAKER'S PERCENTAGES

INGREDIENT	POUNDS	OUNCES	GRAMS	BAKER'S PERCENTAGE
Flour, bread	6		3000	60%
Flour, whole wheat	4		2000	40%
Water	5	10.0	2800	56%
Yeast, compressed		6.0	190	4%
Salt		3.0	95	2%
Total	16	3.0	8085	162%

Note: Metric measures in this table and throughout the text are not necessarily exact conversions of U.S./imperial measures. This is done to avoid the use of awkward numbers. Because baker's percentages remain approximately the same, products are the same, regardless of the units of measure.

TABLE 1.5 ■ DATE FILLING FORMULA EXPRESSED IN WEIGHT AND IN BAKER'S PERCENTAGES

INGREDIENT	POUNDS	GRAMS	BAKER'S PERCENTAGE
Dates	6	3000	100%
Sugar	1	500	17%
Water	3	1500	50%
Total	10	5000	167%

TABLE 1.6 ■ WHOLE WHEAT BREAD FORMULA EXPRESSED IN WEIGHT AND IN PERCENTAGE OF TOTAL BATCH

INGREDIENT	POUNDS	OUNCES	GRAMS	PERCENTAGE OF TOTAL BATCH
Flour, bread	6		3000	37%
Flour, whole wheat	4		2000	25%
Water	5	10.0	2800	35%
Yeast, compressed		6.0	190	2%
Salt		3.0	95	1%
Total	16	3.0	8085	100%